Manual for Convert PDF pages to PNG (s-CPDF2PNG.R)

2026-11-07

© 2026 Peter B. Mandeville.

This work is distributed under a Creative Commons Attribution 4.0 International License (CC BY 4.0).
To view full license terms, visit http://creativecommons.org/licenses/by/4.0/.

Suggested Citation & Attribution:

Mandeville, P. B. (2026). Utilities: Manual for Convert PDF pages to PNG (s-CPDF2PNG.R).

Description:
This **R** utility extracts specified pages from a **PDF** document and renders them as high-resolution **PNG** image files. Utilizing **pdftools** with vectorized page extraction, it provides custom **DPI** resolution controls, defensive page-range validation, and output directory auto-creation.

Instructions for Use:

1. Prerequisites:
Ensure you have **R** installed along with the **pdftools** and **fs** packages:

```
install.packages(c("pdftools", "fs"))
```

2. Configuration & Execution Steps:

1. Configure Script Parameters: Specify input **PDF**, page range, and resolution.
Open **s-PDF2PNG.R** and modify the configuration variables:

- Set **pdf_file** to the target **PDF** file path (e.g., "EDA3-HD9-cad 2026-07-27_12-32-49.pdf").
- List target page numbers in the pages vector (e.g., c(1, 2)).
- Set **out_file_base** for output naming and **out_dir** for the destination folder.
- Adjust **dpi** (default is **300** for publication-quality images).

2. Execute Image Extraction: Run the script in **R** console or **RStudio**.
Execute the script or call extract_plot_pages(...). The console will display rendering status and output locations:

```
Saved PNG: include/EDA3-HD9-cad_page_1.png
Saved PNG: include/EDA3-HD9-cad_page_2.png
==== s-PDF2PNG.R Script Complete ====
```

3. Review Rendered PNGs: Locate generated image files.
Navigate to your configured **out_dir** folder (include/) to access your extracted **PNG** images named according to the <**out_file_base**>_page_<**N**>.png schema.